

Weathering the Artificial Intelligence revolution: public attitudes to potential adoption of Machine Learning-based forecasts.



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ABSTRACT

We present the results from a survey of over 6,000 members of the UK public to understand how they view the potential accuracy of Machine Learning Weather Prediction (MLWP) forecasts, how confident they would feel using this information to make decisions, and how these views compare to their opinions of physics-based Numerical Weather Prediction (NWP) forecasts. Most respondents (~78%) perceive current NWP forecasts as accurate, with only a very small proportion (~1%) indicating that they “Don’t know” the accuracy of these forecasts. This perception is consistent across age, gender, social grade, and geographic regions across the UK. In contrast, there is much less agreement across groups about the potential accuracy of MLWP forecasts: only around 47% of respondents believe these forecasts would be accurate, with roughly a fifth of respondents indicating they “Don’t know” what the accuracy of ML-based forecasts would be. These findings highlight a contrast between current perceptions of NWP, shaped through direct experience, and current perceptions of MLWP, shaped largely by prior beliefs. However, the high agreement for perceived accuracy of physics-based NWP forecasts suggests that users may be willing to update their beliefs about MLWP capabilities, and that direct experience of using the information is likely to be important. The diverse perceptions across demographic, socioeconomic, and regional groups, together with emerging research on AI’s role in weather forecasting and risk communication, underscore the need for broader public engagement. This engagement should not only identify potential concerns, but also inform user-centred solutions that can be embedded within forecast systems to strengthen trust and confidence.

1. Introduction

As the Artificial Intelligence (AI) Revolution (Clark et al., 2019; UKRI, 2021) continues to gain pace, its impact is rippling across all aspects of society - rewriting the rules of entire industries (HM Government, 2021) and fundamentally changing the way we work and live. This is as true in meteorology as it is for any other sector, with recent years bearing witness to a step-change in the performance of machine learning (ML) models for predicting the weather (Keisler 2022, Lam et al. 2022, Pathak et al. 2022, Bi et al. 2023, Chen et al. 2023a, Ben Bouallègue et al. 2024). Machine Learning Weather Prediction (MLWP) offers the promise of increased accuracy and timeliness (de Burgh-Day and Leeuwenburg 2023) and,

while model training costs remain significant, the computational cost of running the trained model, referred to as ‘inference’, is several orders of magnitude less than Numerical Weather Prediction (NWP) (Nipen et al. 2024). With increasing sophistication and improving performance, MLWP models are rapidly approaching the point when they may be used alongside physics-based NWP in operational systems to deliver forecasts for the public.

At the same time, the broader landscape of AI governance is evolving rapidly: governments and regulatory bodies are introducing transparency requirements about AI use, such as the EU AI Act, alongside national policies and standards that seek to define and refine requirements around accountability and the responsible use of AI. Complementary initiatives such as the UK’s Algorithmic Transparency Recording Standard (ATRS) illustrate a broader movement towards documenting and communicating the use of algorithmic tools in within the public sector. In this context, the term “algorithmic tools” is used as a broad descriptor that covers applications of AI, statistical modelling and advanced computational methods. These considerations are also increasingly important for the weather and climate community, as AI capabilities continue to advance, alongside growing expectations for transparency.

Building and maintaining trust in weather forecasts are especially critical for National Meteorological Services, where a lack of user trust could have serious implications for lives and livelihoods, particularly if it undermines the effectiveness of public weather warnings. More broadly, there is growing recognition of the need for equitable weather and climate services that ensure all communities, especially vulnerable groups, can benefit from timely and actionable information. While equity has been more widely explored in climate services (e.g. Doblas-Reyes, et al, 2024), there are strong reasons to incorporate similar thinking into weather forecast services.

Addressing the challenges of trust and equity makes it increasingly important to understand how users, including the public, perceive and respond to MLWP-based forecasts, particularly as such systems are becoming more operationally relevant. However, research on perceptions of MLWP remains limited, with most studies focusing on differences among professional user groups, including forecasters and domain experts. Within these groups, evidence suggests cautious optimism about the potential of AI to enhance weather warnings, tempered by concerns about over-reliance (Kox et al. 2025). Expert users also differ in their priorities, with some preferring data timeliness and availability, and others valuing clarity on

system limitations (Harrison et al. 2025). Interestingly, one study found that labeling products as AI or ML did not reduce trust and, in some cases, increased enthusiasm (Wirz et al. 2024). We note that some of the concerns about MLWP among expert users are reminiscent of scepticism that marked the early development of NWP (Martin-Nielsen, 2017), reinforcing the gradual nature of trust-development, and the importance of aligning system design with user needs (Cain et al. 2024).

A small but growing body of research has also examined the role of AI in weather forecasting and risk communication across diverse user groups. Such work underscores two key points: (i) trust is not a fixed property of technology, but an evolving judgment shaped by user experience and context, even when systems adhere to formal standards (Wirz et al. 2025); and (ii) realising the potential of AI to help mitigate compounding weather-related risks without introducing new ones requires a nuanced understanding of trustworthiness. This includes consideration of decision-making needs, the intrinsic and contextual quality of information, model development processes, and transparency (McGovern et al. 2024).

With the growing ambition to embed AI and ML into public services (HM Government, 2025), there is a clear need to systematically collate and explore public perceptions of these technologies. These efforts should build on the expanding body of research outlined above, ensuring that insights are translated into effective and practical engagement strategies. Doing so will not only support the development of transparency and trust, but will also advance our understanding of public acceptance, and the conditions under which emerging technologies can be responsibly deployed.

To this end, several UK public surveys have benchmarked general attitudes and concerns regarding the use of AI, including the annual ‘Public attitudes to data and AI: Tracker survey’ run by the Centre for Data Ethics and Innovation since 2022. However, to date, no surveys have specifically focused on public perceptions of AI in the context of operational weather forecasting. This study addresses that gap, with the aim of informing further anticipated integration of AI-based methods within public service weather forecasts. In particular, the findings could contribute to an emerging evidence base to inform approaches for the adoption, and effective integration of MLWP alongside NWP.

Achieving these objectives requires the use of clear and consistent terminology to build a shared understanding between the scientific community and the public. However, the

increasing prevalence of technical language in this domain can introduce ambiguity, particularly in public-facing contexts. In public surveys, the term “AI” is more commonly used than “ML,” reflecting its broader and more intuitive framing. To maintain clarity, we use “ML” when referring to the scientific forecasting capability and “AI” in survey questions posed to the public, ensuring technical accuracy and accessibility, while minimising the risk of misinterpretation.

2. Public Survey of accuracy and confidence

Every year, the UK Met Office commissions research to understand consumers’ perceptions of its Public Weather Service (PWS). To assess current public perceptions about MLWP forecast accuracy, and confidence using forecast information for decision-making, we incorporated the following targeted questions in recent PWS surveys, with responses given on a Likert scale:

- 1) Have you ever heard of the term Artificial Intelligence (AI)?
- 2) How accurate, if at all, do you think most weather forecasts are for the following types of weather?
 - a. Daily weather – e.g. sunshine, light rain showers
 - b. Severe weather – e.g. thunderstorms, very strong winds
- 3) Artificial Intelligence (AI) involves using computers and machines to do things that have traditionally been done by humans. This includes creating ways to categorise, analyse, and make predictions from data. AI is currently used in a variety of everyday situations from facial recognition to medical diagnostics tools. Traditional weather forecasts are physics-based, but AI shows significant potential for use in weather forecasting. How accurate, if at all, do you think AI-generated forecasts would be for the following types of weather?
 - a. Daily weather – e.g. sunshine, light rain showers
 - b. Severe weather – e.g. thunderstorms, very strong winds
- 4) How confident, if at all, are you using weather forecasts when making the following types of decisions?

a. Typical daily decisions – those concerning daily tasks such as commuting to work, household chores, also leisure activities such as having a barbecue or taking part in sport/exercise

b. Important decisions – those concerning the impacts of severe weather on you, your property or vulnerable people you care for, your plans for important events such as a wedding, or decisions impacting livelihood (industries such as farming or construction)

5) How confident, if at all, do you think you would be when making these decisions using AI-generated weather forecasts?

a. Typical daily decisions – those concerning daily tasks such as commuting to work, household chores, also leisure activities such as having a barbecue or taking part in sport/exercise

b. Important decisions – those concerning the impacts of severe weather on you, your property or vulnerable people you care for, your plans for important events such as a wedding, or decisions impacting livelihood (industries such as farming or construction)

The surveys were conducted using an online interview administered to members of the YouGov Plc UK panel of 2.5 million+ individuals who have agreed to take part in surveys. E-mails are sent to panelists selected at random from the base sample inviting them to take part in a survey and via a generic survey link. Once a panel member clicks on the link they are directed to the survey that they are most required for, according to the sample definition and quotas which ensure the responses reflect the UK adult population by age, gender, region, and social grade. Invitations to surveys do not expire and respondents can be sent to any available survey.

Following the protocol outlined above, a total of 6,022 UK adults aged 18+ were surveyed in three waves - October 2024 (2,043 people), January 2025 (2,011 people) and April 2025 (1,968 people). Summary statistics show that differences between the sample and national population are minimal, being typically less than 1% for most groups and never exceeding 6%. While the sample is largely representative of the UK population, we want to avoid over-generalising and, therefore, use the raw data, i.e. we did not apply weights which are sometimes used to make the survey data fully representative of the UK adult population.

Clearly, the public are not currently using a machine learning weather forecast, so the public perceptions contrast direct experience of current forecasts (based on NWP), with prior

beliefs about MLWP forecasts based on their knowledge about AI and other factors described above. The survey responses, therefore, provide a baseline against which we can compare changing perceptions as the public becomes more familiar with MLWP. To some extent, the survey questions represent a false dichotomy in that many weather forecasting capabilities of the future will involve a blend of physics-based and machine learning components (Shipway et al., 2026). However, asking separate questions about NWP and MLWP is necessary for understanding prior beliefs, and enables us to explore with more clarity how labelling a forecast as MLWP could influence public perceptions of its accuracy, and their confidence using the information for making decisions. Caution is, therefore, needed when interpreting absolute differences between the public perceptions of NWP and MLWP forecasts, because responses based on personal experience (potentially associated with a human forecaster) may not be directly comparable to perceptions about something that is largely unknown. Perhaps more important are the differing perceptions of NWP and MLWP within each demographic, socioeconomic and regional group, which will be important considerations when engaging with the public about evolving weather forecasting capabilities.

3. Perceptions of accuracy and confidence

The responses to Question 1 (Figure 1) show that more than 95% (5,751/6,022) of respondents had previously heard of AI, comparable with other recent surveys (Public attitudes to data and AI: Tracker survey (Wave 4) report - GOV.UK). Just over 20% (1,291/6,022) reported that they could explain AI in detail, with nearly 60% able to give a partial explanation. In our full dataset (not presented here), self-reported AI knowledge shows differences between demographic, socioeconomic and regional groups, which may affect how MLWP forecasts are perceived and used by different parts of the population (discussed later). Further research is needed to understand specific concerns and how to build trust across these different groups so that everyone can benefit from new capabilities, potentially involving tailored engagement strategies, including further surveys, workshops, and focus groups.

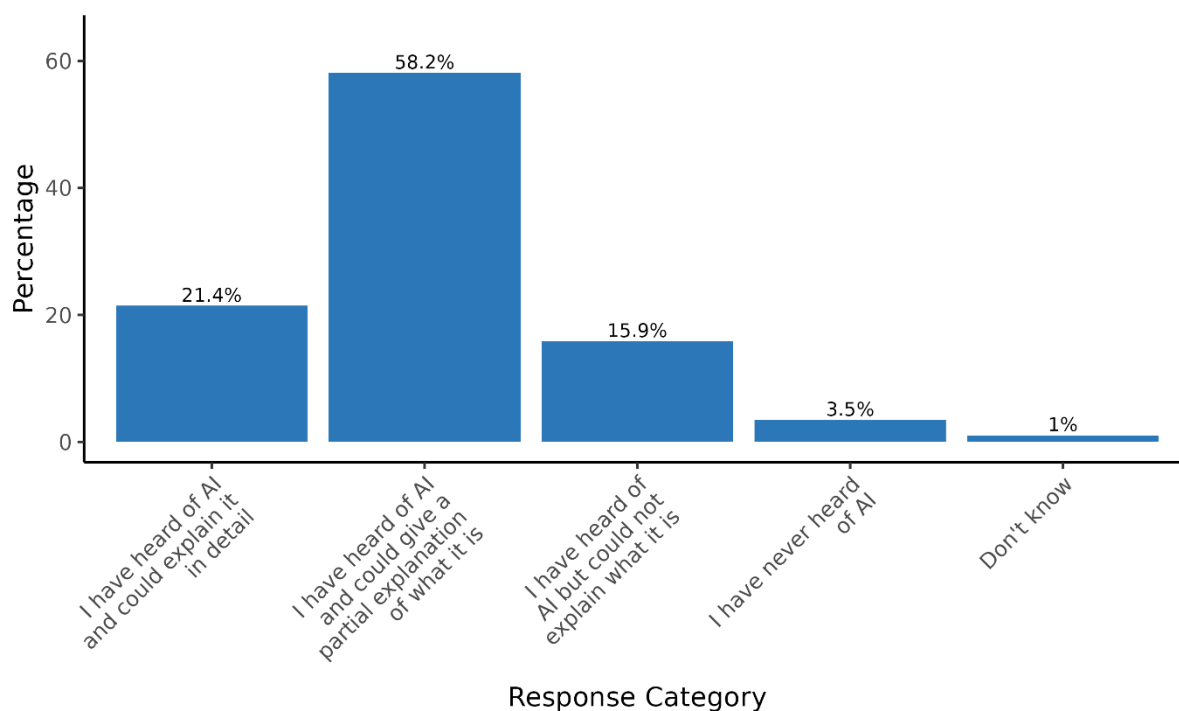


Fig. 1: Self-reported AI knowledge (Question 1: have you ever heard of the term Artificial Intelligence (AI)?).

Public perceptions of NWP and MLWP forecast accuracy are shown in Figure 2. Roughly ~78% of respondents rated NWP forecasts as either “Very accurate” or “Slightly accurate”, compared to 45-48% for MWLP forecasts, both for daily and severe weather. In addition, ~20% of respondents “Don’t know” how accurate MLWP forecasts would be, compared to only 0.5-1% for NWP forecasts. We see a similar pattern in responses about confidence using forecast information for daily and important decisions (Figure 3), with 80-88% of respondents confident using NWP information, compared to 40-44% for MWLP forecasts. Roughly 17% of respondents “Don’t know” how confident they would be using MLWP forecasts, compared to 2-4% for NWP. In each case, the more even spread in responses about MLWP forecast accuracy and confidence using the information suggests a degree of rational scepticism, i.e. in the absence of direct experience, the public is yet to reach a consensus about potential AI forecasting capabilities and how to use them.

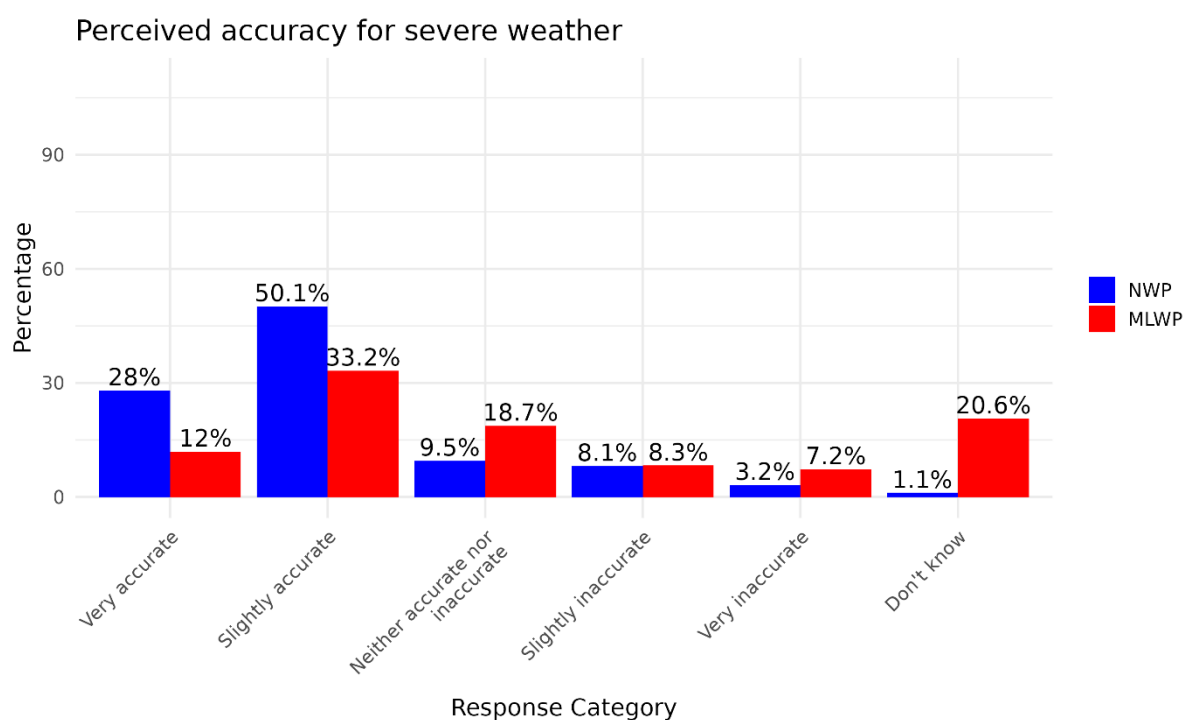
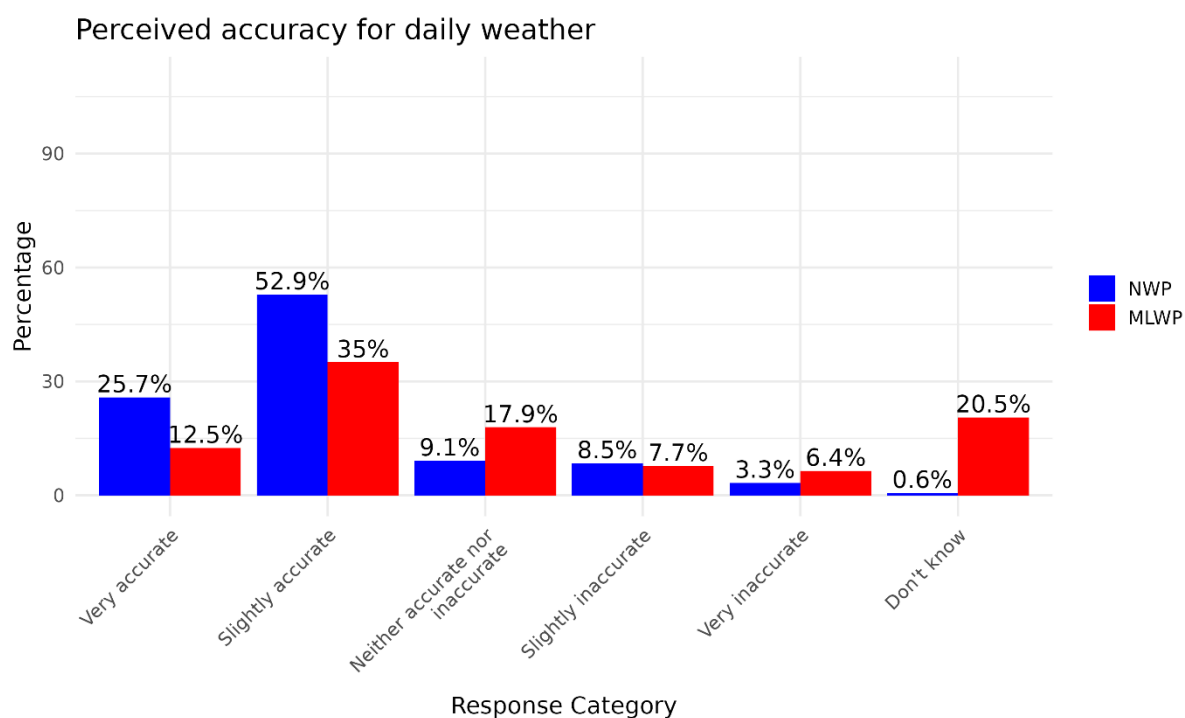


Fig. 2: Perceptions of forecast accuracy for NWP (Question 2; blue) and MLWP (Question 3; red) for typical daily weather (a; upper panel) and severe weather (b; lower panel).

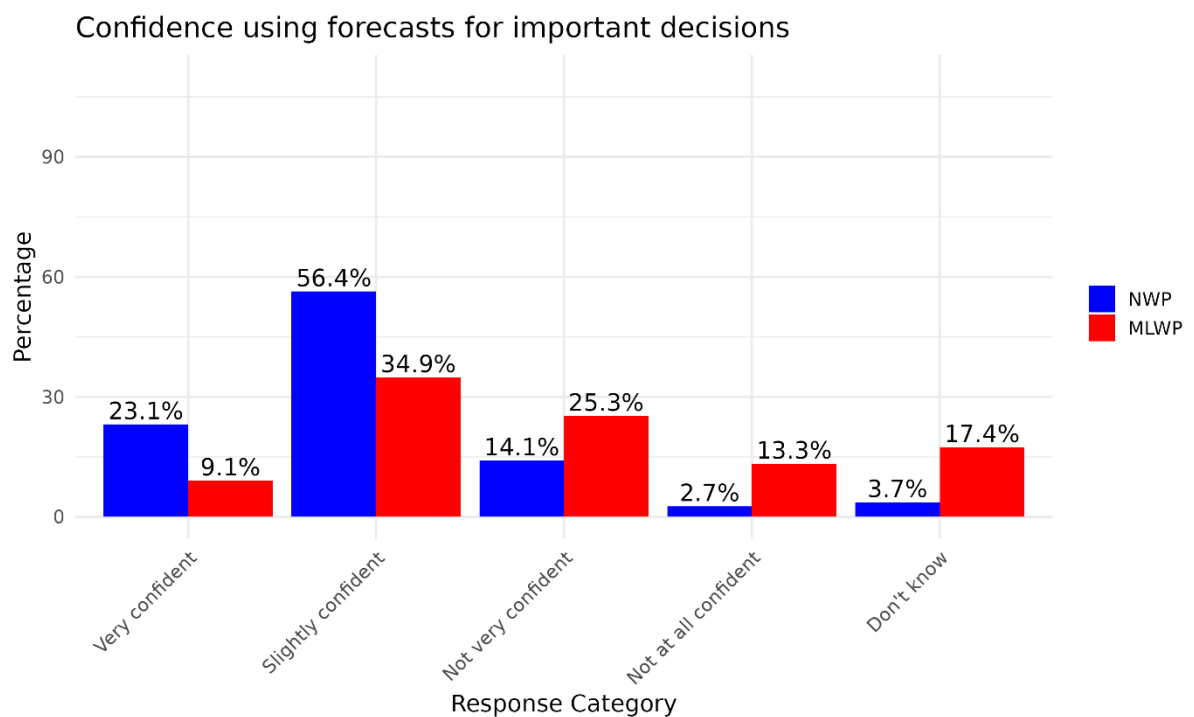
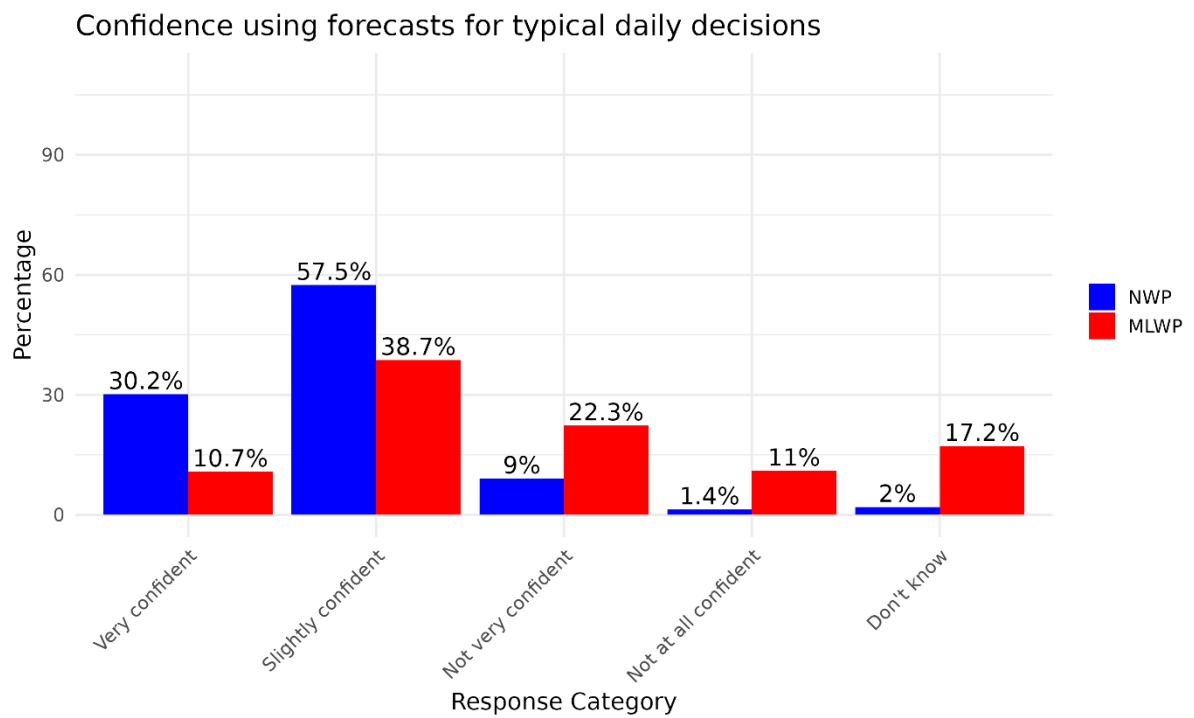


Fig. 3: Confidence using NWP weather forecasts (Question 4; blue) and MLWP (Question 5; red) for typical daily decisions (a; upper panel) and important decisions (b; lower panel).

The processed survey data provided to the Met Office are organised into blocks that represent demographic, socioeconomic and regional groups, as well as self-reported AI knowledge. As shown in the top-row of Table 1, each block contains between 2 groups (e.g. gender) and 13 groups (e.g. region), enabling us to explore how perceptions vary between groups within a block (for example, their dependence on age). However, because the processed survey data provided by YouGov are aggregated at the group level, individual responses are not available, and it is not possible to explore how perceptions vary simultaneously across multiple characteristics. For this reason, the analysis is limited to summarising the top-level findings for each block, noting that there will be latent factors that influence perceptions which this survey does not capture.

To illustrate these patterns, Table 1 summarises the groups with the highest and lowest fraction of respondents who perceive forecasts as accurate and who would be confident using them for both NWP and MLWP. The corresponding numerical values are presented in Figure 4, where they are represented by the endpoints of the connected dots (often referred to as “dumbbells”). For example, in response to Question 2a, the South East regional group had the highest fraction of respondents who perceived NWP forecasts as accurate, while Wales had the lowest fraction. Similarly, the 25–34-year-old age group had the highest fraction of respondents who perceived both NWP and MLWP forecasts as accurate, whereas the 35–44 age group had the lowest fraction.

Blocks	Gender	Region	Urban/ Rural	Social Grade	Age (years)	AI knowledge
Groups	Male; Female	North East; North West; Yorkshire and	Urban; Town and	ABC1, C2DE ¹	18-24; 25- 34; 35-44;	Heard of AI, could explain in detail; Heard of AI, could

1

<https://www.ons.gov.uk/census/aboutcensus/censusproducts/approximatedsocialgradedata>

		the Humber; East Midlands; West Midlands; East of England; London; South East; South West; England; Wales; Scotland; Northern Ireland	Fringe; Rural		45-54; 55-64; 65+	explain partially; Heard of AI, could not explain; Not heard of AI
Q2a. NWP accuracy (daily)	Max = Female; Min = Male	Max = South East; Min = Wales	Max = Town and Fringe; Min = Rural	Max = ABC1; Min = C2DE	Max = 25-34; Min = 35-44	Max = Heard of AI, could explain; Min = Heard of AI, could not explain
Q3a. MLWP accuracy (daily)	Max = Male; Min = Female	Max = London; Min = North East	Max = Urban; Min = Rural	Max = ABC1; Min = C2DE	Max = 25-34; Min = 65+	Max = Heard of AI, could explain; Min = Heard of AI, could not explain
Q2b. NWP accuracy (severe)	Max = Female; Min = Male	Max = Northern Ireland; Min = North East	Max = Urban; Min = Rural	Max = ABC1; Min = C2DE	Max = 65+; Min = 55-64	Max = Heard of AI, could explain; Min = Heard of AI, could not explain
Q3b. MLWP accuracy (severe)	Max = Male; Min = Female	Max = London; Min = Scotland	Max = Urban; Min = Town and Fringe	Max = ABC1; Min = C2DE	Max = 25-34; Min = 55-64	Max = Heard of AI, could explain; Min = Heard of AI, could not explain
Q4a. NWP confidence (typical)	Max = Female; Min = Male	Max = East of England; Min = North East	Max = Rural; Min = Urban	Max = ABC1; Min = C2DE	Max = 25-34; Min = 18-24	Max = Heard of AI, could explain; Min = Heard of AI, could not explain
Q5a. MLWP	Max = Male; Min = Female	Max = London; Min = Scotland	Max = Urban; Min =	Max = ABC1; Min = C2DE	Max = 25-34; Min = 45-54	Max = Heard of AI, could explain; Min = Heard of AI, could not explain

confidence (typical)			Town and Fringe			
Q4b. NWP confidence (important)	Max = Male; Min = Female	Max = North West; Min = Yorkshire and the Humber	Max = Town and Fringe; Min = Rural	Max = ABC1; Min = C2DE	Max = 45-54; Min = 18-24	Max = Heard of AI, could explain; Min = Heard of AI, could not explain
Q5b. MLWP confidence (important)	Max = Male; Min = Female	Max = London; Min = Scotland	Max = Urban; Min = Town and Fringe	Max = ABC1; Min = C2DE	Max = 25-34; Min = 45-54	Max = Heard of AI, could explain; Min = Heard of AI, could not explain

Table 1: Summary of survey responses for questions 2-5, by group within each block. For both NWP and MLWP forecasts, the table highlights the groups with the highest (max) and lowest (min) fraction of respondents who perceive the forecasts as accurate and who would be confident using them. The top row lists the groups within each block. Figure 4 shows the numerical values corresponding to these groups, where they are represented by the endpoints of the connected dots (“dumbbells”).

To demonstrate how perceptions vary within and between each block, Figure 4 shows a subset of the quantitative responses about perceived accuracy and confidence for each group within the demographic, socioeconomic, regional and AI knowledge blocks. For clarity, we simplify the Likert-scale responses using the following rules:

- Accuracy: for each block and group, respondents are categorised into those who rated the forecast as “Accurate” (“Very” or “Slightly accurate”), (ii) rated it as “Inaccurate” (“Slightly” or “Very inaccurate”), and (iii) selected “Don’t know”.
- Confidence: for each block and group, respondents are categorised into those who were “Confident” (“Very” or “Slightly confident”), (ii) “Not confident” (“Slightly” or “Very unconfident”), and (iii) those who selected “Don’t know”.

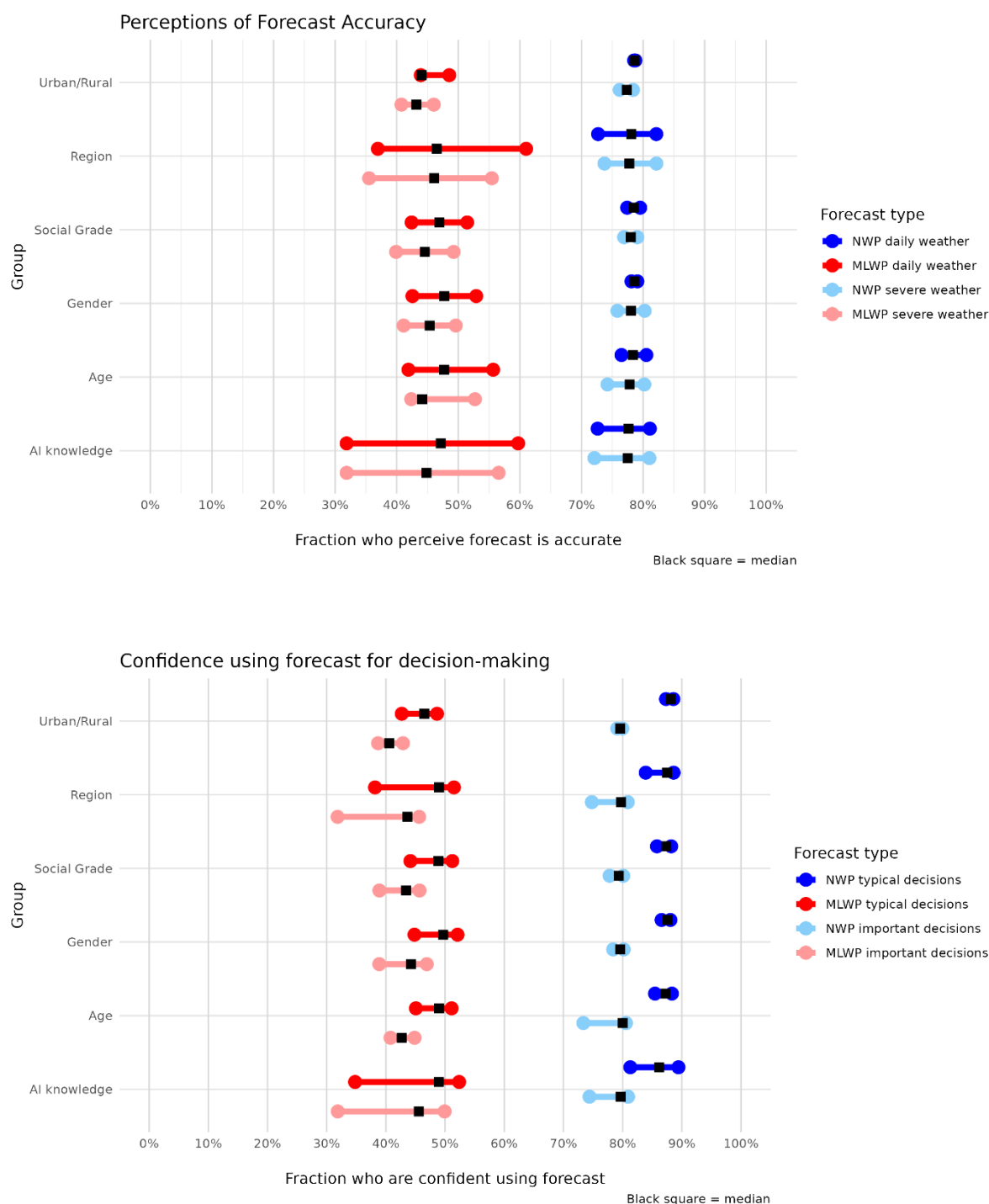


Figure 4: Connected dot (“dumbbell”) plots showing fraction of respondents to perceive the forecast as accurate (upper panel) and the fraction who are confident using forecast information for decision-making (lower panel) within each response block. Each “dumbbell” represents the range of responses by showing the lowest (min) and highest (max) fraction for each block, with the median fraction shown in grey. Blue colours show results for NWP, red for MLWP. The darker shades show

the responses for daily weather and typical decisions, and the lighter shades show the responses for severe weather and important decisions, respectively. The ends of each dumbbell correspond to the groups summarised in Table 1.

The fraction of respondents for each block and group who perceive the forecast as accurate (upper panel), and who would be confident using forecast information for decision-making (lower panel) are shown in Figure 4. Each dumbbell represents the range of responses within the group, showing the minimum and maximum fraction (corresponding to the groups listed in Table 1), with the median response shown in grey. Blue represents NWP, and red represents MLWP, with darker shades showing the responses for daily weather and typical daily decisions, while the lighter shades show the responses for severe weather and important decisions.

There is remarkable agreement across demographic, socioeconomic and regional groups in their perceptions of NWP forecast accuracy. This is probably because NWP is a well-established approach, and users have had many years of direct experience using the information for decision-making, i.e. they understand what it does well, what it does not, and how to use it. In contrast, AI is an emerging technology and, in the absence of direct experience, a smaller fraction of each block perceives it to be accurate, when compared to traditional NWP forecasts.

In addition to systematic differences in blockwise perceptions of forecast accuracy, there is a much wider spread in responses about MLWP forecasts across demographic, socioeconomic and regional groups about its potential accuracy. However, given the small number of groups within each block, formal statistical inference and significance tests are not robust. For that reason, we do not calculate the statistical significance of differences in the perceptions of accuracy between MLWP and NWP. Instead, we adopt a consistency-based approach. Across all blocks, the median fraction who perceive the forecast as accurate is always lower for MLWP than for NWP. Under the null hypothesis, there is no systematic difference in perceived accuracy between MLWP and NWP forecasts. It follows that, for any given block, the probability that the median perceived accuracy for MLWP is lower than that for NWP is 0.5, reflecting random variation with no directional bias. In the observed data, this occurs in all six blocks. Assuming independence across blocks, the probability of this outcome arising by chance is $0.5^6 \approx 0.016$. This low probability provides evidence of a

systematic difference in perceptions about the two forecast types, with respondents currently rating MLWP forecasts as less accurate than NWP forecasts. The same pattern is observed in the range of responses: the difference between the maximum and minimum fractions is always greater for MLWP than NWP, indicating greater variability in perceptions of MLWP accuracy.

The results are broadly similar to the above for responses about confidence in using forecast information for decision-making. A slightly higher fraction of people report being confident using NWP forecasts to inform daily decisions than perceiving them as accurate. If meaningful, this could suggest that some individuals who think the forecast is inaccurate are still confident using it. However, it is also possible that this difference arises from variations in the response categories between accuracy and confidence questions; as such, this interpretation should be treated with caution.

Consistently, the median fraction of respondents who would be confident using the forecast is always lower for MLWP than for NWP. In addition, the range of responses (difference between the maximum and minimum fractions) is greater for MLWP than NWP in five out of six blocks. Under the null hypothesis of no systematic variability between forecast types, MLWP would be expected to exhibit a larger spread than NWP in any given block with probability 0.5. Assuming independence across blocks, the probability of observing this pattern in five out of six blocks by chance is $0.5^5 \approx 0.03$. The low probability suggests that the pattern is unlikely to arise from random variation alone, providing some evidence that perceived accuracy and confidence vary more for MLWP than for NWP. More generally, the distribution of responses for NWP provides a useful baseline against which changes in perceptions of MLWP can be assessed in future surveys.

As described above, Table 1 summarises the groups within each block that exhibit the highest and lowest fractions of respondents who perceive forecasts as accurate and who report confidence using them. This highlights several consistent patterns across both NWP and MLWP, alongside some notable differences:

- **AI knowledge** – respondents who report having heard of AI and being able to explain it show the highest fractions perceiving both NWP and MLWP forecasts as accurate and who would be confident using them. In contrast, those who have heard of AI and cannot explain it consistently show the lowest fractions perceiving NWP and MLWP forecasts as accurate, and confidence in using them.

- **Social grade** – respondents in the ABC1 group have higher fractions than those in the C2DE group who perceive both NWP and MLWP forecasts as accurate, and would be confident using them.
- **Gender** – a higher fraction of women than men perceive NWP forecasts as accurate, whereas a higher fraction of men than women perceive MLWP forecasts as accurate. The responses for confidence are less consistent.
- **Region** – in general, but not uniformly, respondents in the South, East, and London show higher fractions perceiving NWP and MLWP forecasts as accurate and who would be confident using them. Those in the North East, Scotland, and Wales groups tend to have the lowest fractions perceiving NWP and MLWP forecasts as accurate and who would be confident using them.
- **Age** – the 25-34 group most frequently exhibits the highest fraction of respondents who perceive both NWP and MLWP forecasts as accurate and who would be confident using them.
- **Urban/Rural** – there is limited evidence of a systematic pattern across these groups; however, within this block, the rural respondents most frequently exhibit the lowest fractions perceiving the NWP and MLWP forecast as accurate, and who would be confident using them.

These patterns are largely consistent with established findings on the UK's digital divide (ONS, 2019), despite differences in survey design. While caution is warranted in interpreting the findings, they suggest a risk of reinforcing existing disparities in engagement with emerging technologies. This further highlights the importance of carefully considering the needs of different user groups to ensure equitable access and build understanding of new forecasting approaches. Addressing these factors may help align public perceptions of MLWP with those currently associated with NWP, both in terms of perceived accuracy and confidence using the information.

4. Bridging the gap

The observed gap between perceptions of NWP and MLWP seems to highlight the need to build wider public trust in weather forecasting approaches that include machine learning.

However, the results also highlight the limits of what we can learn from surveys alone. In particular, the substantial minority of respondents (approximately 17-20%) who indicated that they “Don’t know” how accurate MLWP forecasts would be, or how confident they would be using them for decision-making, warrants close attention. This group may reflect uncertainty, unfamiliarity or ambivalence rather than a neutral evaluation, and, therefore, represents an important target for further research and engagement.

Understanding the basis for these views and the reasons behind them will be important because many, if not most, public users are unlikely to interrogate formal forecast performance metrics that are familiar to model developers. Indeed, previous work exploring trust in AI system (e.g. McGovern et al 2024; Wirtz et al 2025) suggests that users will want to evaluate themselves whether, when, and to what extent AI outputs can be trusted, even when they have been developed according to formal technical standards. In this context, transparency, particularly in how forecasts are generated, communicated and interpreted, will be central to supporting understanding and trust, and aligns with broader regulatory developments and emerging best practice.

Taken together, these findings point to the need for complementary research that extends beyond structured surveys. Qualitative methods, including focus groups, interviews and workshops, could help identify the reasons underpinning both negative and uncertain perceptions of MLWP, and clarify what aspects raise concern, confusion, or scepticism. Such mixed-method approaches would provide valuable context for interpreting survey responses, and for informing effective communication strategies, public engagement, and building trust.

How quickly perceptions will change is unclear, with service providers having a vital role to play. In a general description of this problem, Pope et al. (2017) proposed a Bayesian decision-theoretic model in which perceived forecast value depended on both the user’s prior beliefs about forecast accuracy, and evidence of forecast performance from direct experience. In that model, perceptions of accuracy generally changed more slowly for users who were sceptical, i.e. those who either under-estimated or were very uncertain about the forecast accuracy - as is the case for MLWP. There could be many reasons for different prior beliefs, including trust in institutions that produce forecasts, previous experience, as well as perceived risks and benefits of new technologies. Given this framework, our survey findings suggest prior beliefs vary considerably across demographic, socioeconomic, and regional groups. This, in turn, suggests the need for both early and extensive engagement with the public,

including partnerships with experts in other disciplines (e.g. social sciences), to elicit deeper insights about attitudes towards emerging technologies, and priorities across different groups. Once armed with this information, we encourage the weather forecasting community to work together in ensuring these considerations are adequately represented in the development of MLWP forecast products and services, and ensure both that the digital divide is not reinforced or widened, and that trust in NWP (the backbone of most weather forecast systems) is not undermined. This approach has the potential to pave the way for more rapid realisation of anticipated benefits of integrating NWP and MLWP capabilities (e.g. improving forecast accuracy, timeliness and resolution) to help people make better weather-related decisions.

The public journey towards increasing AI integration in public services, such as weather forecasts, must also extend beyond a requirement for simple audience engagement, and explore a deeper embedding of user insights within technical developments. We are therefore motivated to interpret the survey responses within the context of key science/strategy implications that are relevant to weather forecast providers (including National Meteorological Services), for example:

1. **Approach.** While the public survey treated NWP and MLWP as entirely separate approaches, the choice does not have to be as stark; indeed, data science techniques have long been used within the weather forecast value chain (albeit not necessarily acknowledged under the ML label). However, the gap in perceptions of accuracy and confidence between NWP and MLWP supports the need for continued efforts to combine their respective strengths (e.g. Shipway et al., 2026). Products and services delivered using an integrated approach to NWP and MLWP (which can be very carefully curated and controlled) may help support a responsible approach to AI adoption.

2. **Experience.** As in any community, there are inevitably early-adopters with an appetite for exploiting emerging technologies that both champion the new capability and contribute informed feedback. Releasing experimental code and data to these early-adopters can be highly beneficial. Examples of related initiatives include the advance publication of early modelling work undertaken by Google regarding the GraphCast model (Lam et al 2023) and the exposure of a set of basic data from five MLWP models (AIFS; Aurora; FourCastNet; GraphCast and Pangu-Weather) by the European Centre for Medium-Range Weather Forecast (ECMWF) to enable comparison against its equivalent operational NWP

output (with the UK Met Office doing similarly for the experimental global Artificial Intelligence-based weather forecast system that it co-developed with The Alan Turing Institute, FastNet, e.g. Daub et al., 2025; Dunstan et al., 2025). Additionally, experiments with consenting participants that compare NWP and MLWP, ideally in a context-based decision-making scenario, could yield important insights. Other context-based approaches could include industry-focused collaborative prototyping developments to better understand forecast utility and encourage innovation (e.g. Chen et al., 2023b; Steele et al. 2024; Steele et al., 2025).

3. Evaluation. Subjective measures such as perceived ‘accuracy’ and ‘confidence’ have a natural analogy with standard weather forecast validation/verification measures of ‘skill’ and ‘value’. As such, impartial demonstrations of skill, representation of physical phenomena, and value for decision-making can inform subjective perceptions of accuracy and confidence. Incorporating user-relevant metrics within ML training loss functions and/or forecast performance statistics would help strengthen the design of models and assess their quality in a more-timely manner. There are further opportunities to bring together service developer and user perspectives using argument-based assurance methods, e.g. using the framework proposed by Burr & Leslie (2022). Such assurance frameworks are anticipated to become increasingly important tools for breaking down silos and connecting research and user communities.

Ultimately, even with a deeper embedding of user insights within technical developments, regular public surveys will be required to track progress in the perceptions of NWP and MLWP as the latter becomes more widely used and integrated with NWP.

In benchmarking perceptions of MLWP against NWP it is possible, and indeed probable, that many of the learnings about approaches for building trust could transfer between the two, in much the same way as perceptions of accuracy for forecasting daily weather inform perceptions of severe weather. However, we suspect that this transfer is unlikely to hold across very different timescales and with increasing uncertainty, with users of climate information limited by much slower feedback about how to use it effectively, than for weather forecasts. Thus, while trust in weather forecasts can be strongly influenced by evidence gained through regular, direct experience, building trust in climate information probably requires a stronger emphasis on evidence that informs the user’s prior beliefs, which may include causal explanations described using established mechanistic principles (e.g.

Newton's laws, thermodynamics, etc.). This, of course, is yet to be tested, particularly in the context of AI climate modelling.

Similarly, it is important to recognise that the results presented here are drawn from representative surveys of the UK public. We encourage other service providers to undertake similar surveys, which may reveal contrasting results, particularly in countries where training, skills and experience of AI are very different to the UK – the analysis of which may, in turn, stimulate further ideas for increasing appropriate adoption more generally.

This early foray into public perceptions of AI weather forecasting scratches the surface. The authors hope that it stimulates the work and analysis needed to ensure a smooth transition as MLWP capabilities are integrated alongside NWP. Together MLWP and NWP have the potential to offer the public better, faster, more accurate forecasts, but these will only deliver their true value if they are trusted and used to improve weather-related decision-making.

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